MILITARY SPECIFICATION

WARHEAD SECTION, 762-MILLIMETER ROCKET, CHEMICAL AGENT, M190, WITHOUT FUZE

1. SCOPE

1.1 This specification covers a filled warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

ML-P-116 - Preservation, Methods of.
MIL-T-704 - Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY EDGEWOOD ARSENAL

DL90-11-8 - Warhead Section, 762-MM Rocket, Chemical Agent, Ml 90, W/O Fuze.

(All drawings referenced in this specification are listed on the above drawing.)

FSC 1340

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US ARMY PICATINNY ARSENAL

D8821481 - Ring, Support Assembly.
JS839515 - Shipping and Storage Container, Atomic Warhead Section XM480, E2.
F88339594 - Base, Frame and Suspension Frame Assembly.
F8887952 - Packaging for Warhead Section, 762 MM Rocket, Chemical Agent, M190 and Fuze, M421.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Regulations are a part of the Code of Federal Regulations avaiable from the Superintendent of Documents, Government Printing Office, Washinghm, D. C. 20504. Orders for the above publications should cite "47 CFR 71-90. "

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain that all materials and components prior to their use have met their respective specification.

3.2 Loading and assembly. The waxhead section shall be loaded and assembled as shown o.n Drawing 1190-11-8.

3.3 Leakage. The warhead section shall not show a pressure drop of more than 1 pound per square inch gage (psig) when tested in accordance with 4. 3.4.1.

3.4 Weight. The loaded and assembled warhead section without fuze shall weigh, 1235 /12 pounds (the M421 fuze weighs 3 pounds).

3.5 Longitudinal center of gravity. The longitudinal center of gravity of the loaded and assembled warhead section, including the weight of the M421 fuze, shall be located at station 79. 6/ 1.2 inches when tested in accordance with 4.3.4.2.

3.6 RadiaL center of gravity. The radial center of gravity of the loaded and assembled warhead section shall be within 0.125 inch of its axis when tested in accordance with 4.3,4. 2.

3.7 Preproduction. Prior to the start of regular production, a preproduction sample of warhead sections shall be produced for examination and test in accordance with this specification. (see 4.2).

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3.8 Workmanship. The warhead section shall be free from foreign matter and visible damage such as dents and cracks which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed :necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Lotting. Prior to the start of regular production, two warhead sections shall be produced using the same methods, materials and equipment that will be used in regular production.

4.2.2 Examination and test. The preproduction warheads shall be examined and tested for all the requirements of this specification.

4.2.3 Acceptance/rejection criteria. The preproduction warheads shall meet all the requirements in this specification to be acceptable. The supplier shall obtain written approval from the contracting officer prior to preceding with regular production.

4.3 Regular production.

4.3,1 Lotting. Each warhead section shall utilize the empty warhead section serial number.. No more than two lots of bombs shall be represented in any fully loaded warhead section.

4.3.2 Inspection provisions for examination and test. Each warhead and pack shall be examined and tested in accordance with the list of defects. The acceptance number is 0.

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4.3.3 -List of defects.

4.3.3.1 Warhead section, 762-mm, rocket, chemical agent, M1 90, w/o fuze Dwg D90-11-8.

1. Configuration incorrect (E123-1-293)
2, Leakage (4. 3.4.1)
3. Component missing or incorrectly assembled
4. Detonating cord not sealed
5. Bomb not tightly packed
6. Nose cone not securely fastened
7. Gap at assembly joints excessive
8. Longitudinal center of gravity incorrect (4.3.4.2)
9. Radial center of gravity incorrect (4,3.4.2)
10. Marking incorrect or illegible
11. Workmanship (3. 8)

4.3.3.2 Preparation for delivery.

1. Damaged portions of painted areas on shipping container not treated and repainted as specified
2. Bag fabrication and closure incorrect
3. Air-sampling ports missing, incorrect or inproperly installed
4. Cushioning missing where specified
5. Canister assembly missing or incorrectly installed
6. Warhead section not bolted to ring support assembly
7. Bag leakage
8. Container closure incorrect
9. Container damaged (bent, twisted or punctured)
10. Gasketing of bag not as specified .
11, Ring support assembly not locked in position
12. Marking incorrect, missing or illegible
13. Fuze missing or not secured

4.3.4 Tests shall be conducted as follows:

4.3.4.1 Leakage. Prior to loading, subject the warhead to 5 /1/2 psig internal air pressure for 3 to 4 minutes. After the pressure is stabilized, shut off the air supply prior to observing the gage.

4.3.4.2 Center of gravity. Place the loaded warhead section, with a dummy fuze in position, in test device Q11 6 when examining for the longitudinal center of gravity and in test device Q128 when examining for the radial center of gravity (6.4).

5. PREPARATION FOR DELIVERY

5.1 Preservation, level A,

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5. 1.1 Repainting. Damaged portions 0[painted areas on the steel shipping case and component parts thereto shall be treated and repainted in accordance with MIL-T-704, type A, except that the color of the finish coat of paint shall conform to the requirements of the applicable drawing.

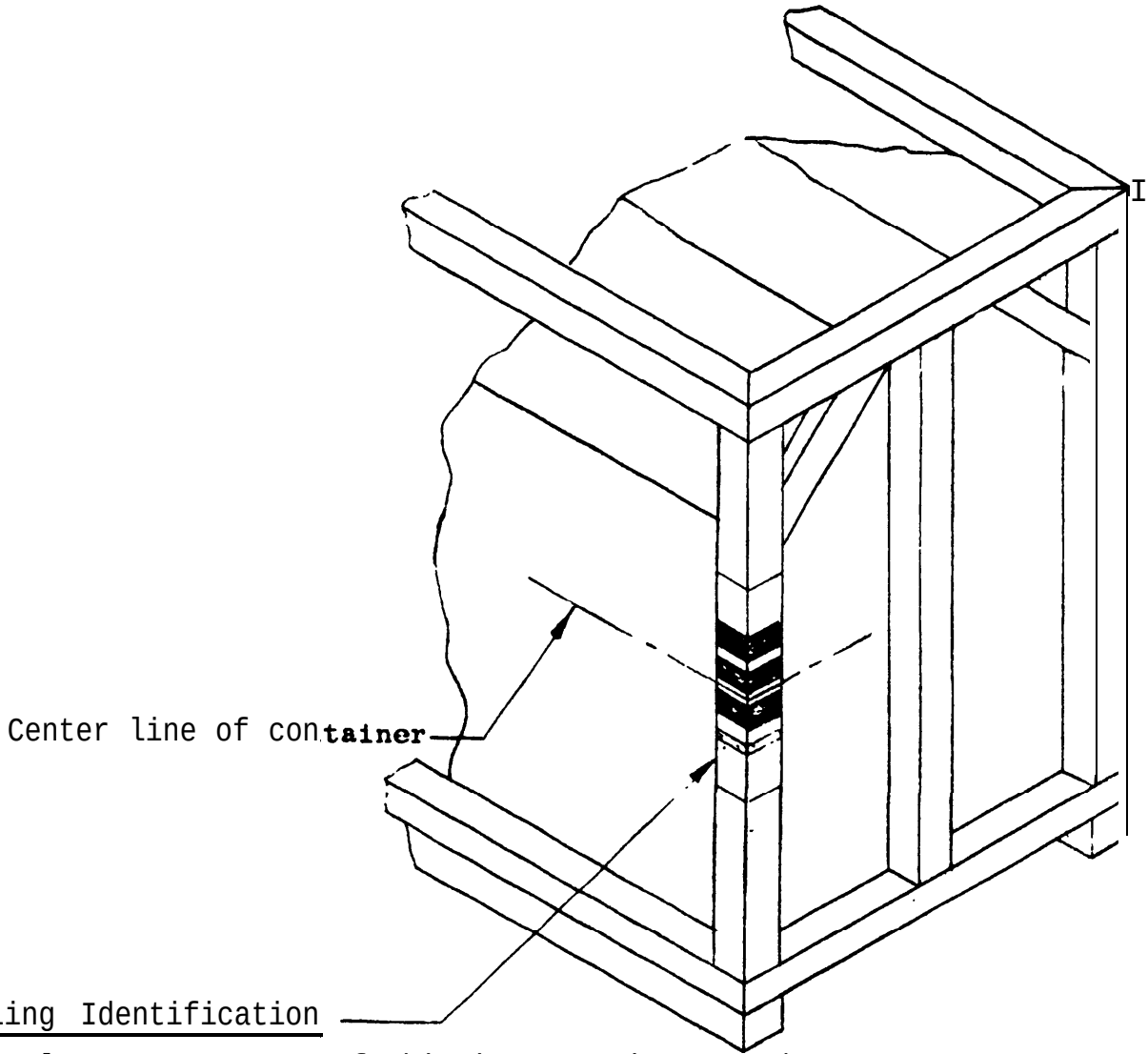
3.2 Packing, level A. The filled warhead section shall be packed in a reusable container conforming to Dwg J8839515. Prior to placing in the container, the filled warhead shall be inclosed in a barrier bag shown on Dwg D90-6-47, which shall be fabricated as shown on Dwg E90-6-48. The warhead shall be lifted in a horizontal position approximately four feet from ground level. Four studs are then inserted into the four aft mounting holes of the warhead with four gaskets placed on the stud bolts (see Dwg D90-6-47). The barrier bag is then placed around the warhead and the four aft holes aligned and positioned over the four stud bolts with the unsealed seam opening in an upward position, The four remaining gaskets and washers are then installed (see Dwg D90-6-47). Before the barrier bag is placed around the warhead and before the warhead is placed in the suspension frame, the area of contact between the barrier material and the warhead, and between the barrier material and the strap assembly and cradle shall be cushioned with 4-inch wide by 1/4 inch thick foam rubber or cellular plastic foam cushioning material. The filled warhead with barrier bag partially installed thereon shall then be lowered into the suspension frame assembly (Dwg F8839594) aligning the warhead in such manner as to allow the four stud bolts to pass through the holes in the assembly ring (Dwg D8821481) on the suspension frame assembly. Caution should be exercised against puncturing or tearing the barrier bag during this operation. The barrier material shall be smoothed out to insure that no wrinkles or creases occur between the warhead and the suspension frame cradle, Bolting of the warhead to the support ring is accomplished by first removing one of the studs and replacing it with a bolt and washer, tightening same to pull the warhead securely against the support ring boss, after which the remaining studs shall be replaced one at a time (see Dwg D90-6-47). The barrier bag shall then have all creases and folds smoothed out as much as possible, after which the bag shall be closed by heat sealing with minimum 1 inch wide seams. To insure a properly sealed bag, the vacuum retention test in MIL-P-116 shall be performed. The filter port outlet plus one of the sampling port outlets should be used to conduct the test. The canister assembly (Dwg C90 -6-89) shall be installed and secured to the base frame assembly within the reusable steel case in the manner as shown on Dwg D90-6-47. Before the reusable steel case is closed, one packaged M421 rocket fuze, shall be secured to the container in the metal box. Attached additional packing and closure procedure shall be in accordance with Dwg F8887952.

5.3 Marking. In addition to any special marking required by the contract or order, the reusable shipping container shall be marked as required in ICC regulation and in accordance with MIL-STD-129,except that color code filling identification shall be as shown on figure 1.

6. NOTES

6.1 Intended use. This item is a component of a chemical rocket.

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Mark alternate corners of Shipping Container as shown. Three l\2-inch wide stripes, color Green No. 24108 or 34108 and one l\4-inch wide stripe, Yellow No. 23538 or 33538, spaced l\2-inch apart on a 4-l/2-inch wide background color Gray No. 26231 or 36231.

Figure 1 - Shipping Container Marking

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for suppliers submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required by the Government to notify the suppiier whether or not to proceed with production.

6.3 Precaution. In view of the hazards attending the handling of agent-filled munitions, suppliers are advised to exercise all necessary precautions to protect personnel from injury during loading of warhead sections.

6.4 Radial center of gravity. Weights (Dwg C123-1-151) may be atttached to the webs at the base of the warhead section to adjust the radial center of gravity. Total weight added should not be enough to cause rejection under 3.4.

6.5 Component specifications. The following specifications reflect the components re quired to complete this item:

MIL-W-46611 - Warhead section, 762-MM Rocket, Empty, C4.
MIL-B-60073 - Bomb, Non-persistent Gas, Spherical, 4. 5-inch, Ml 39; Filling and Assembly.
MKL-B-60075 - Bomb, Non-persistent Gas, Spherical, 4. 5-Inch, Ml 39; Metal Parts.
MIL-B-60076 - Burster, M45.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1340-A209

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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TO DETACH FORM - CUT ON THIS LINE

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